F172 Lesson 9: Gene Therapy Vector Delivery

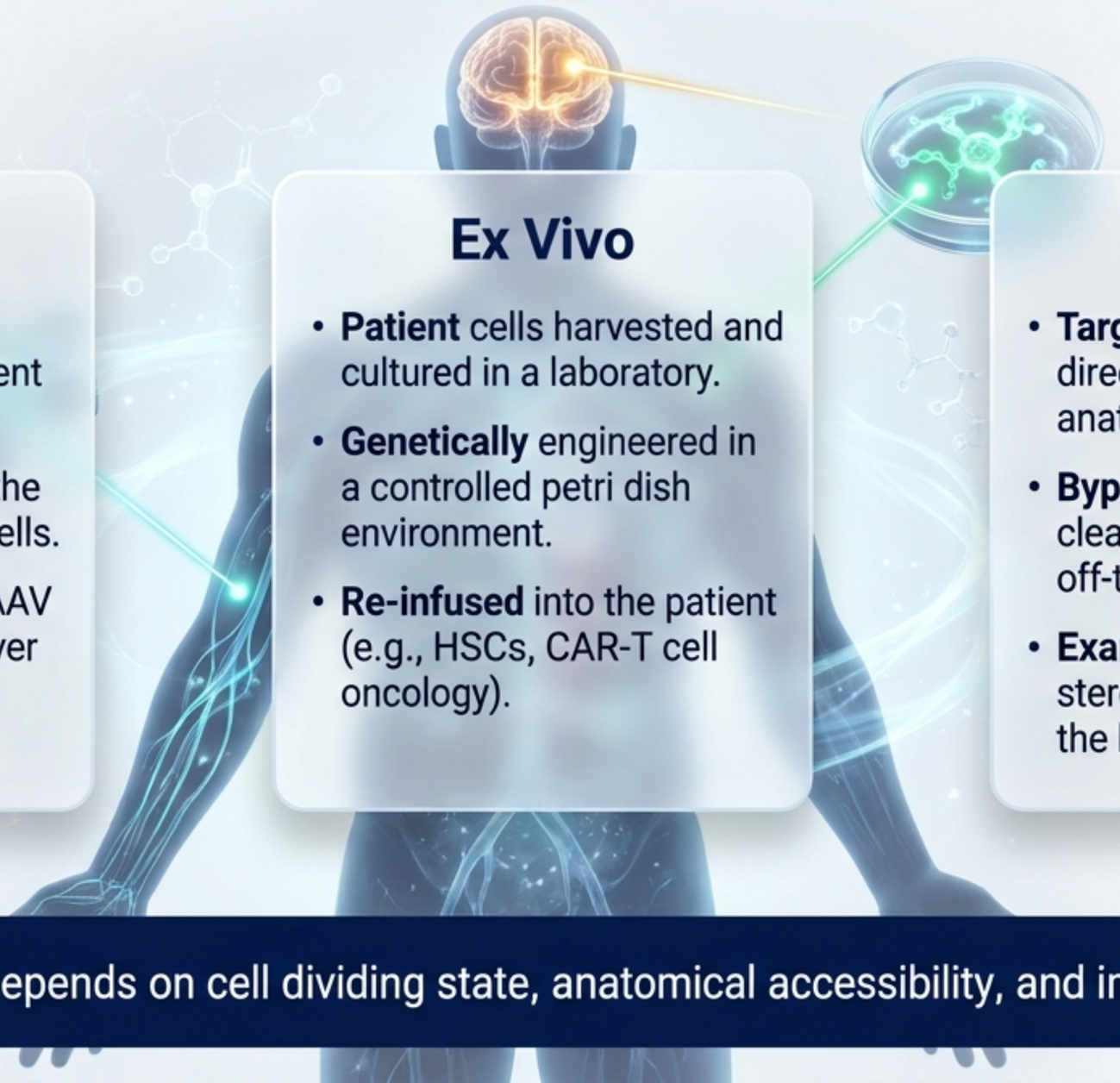
OCR AAQ Level 3 Human Biology · Unit F172: Genetics (Dan)

Viral Vectors | Non-Viral Systems | Anatomical Barriers | Delivery Routes

Differentiate molecular mechanisms, packaging capacities, and clinical trade-offs of viral and non-viral gene delivery systems.

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Delivery Strategies: In Vivo, Ex Vivo & In Situ



In Vivo

- **Direct** systemic administration into patient tissues.
- **Vector** travels through the bloodstream to target cells.
- **Example:** Intravenous AAV infusion targeting the liver or muscles.

Ex Vivo

- **Patient** cells harvested and cultured in a laboratory.
- **Genetically** engineered in a controlled petri dish environment.
- **Re-infused** into the patient (e.g., HSCs, CAR-T cell oncology).

In Situ

- **Targeted** administration directly into a specific anatomical compartment.
- **Bypasses** systemic clearance and reduces off-target effects.
- **Example:** Intrathecal or stereotactic injection into the brain.

Key Principle: Strategy depends on cell dividing state, anatomical accessibility, and immune clearance risks.

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Adeno-Associated Virus (AAV): The CNS Gold Standard

[Vector: Viral] [Integration: Episomal] [Payload: 4.7 kb]

Tropism & Serotypes	• High cellular affinity for non-dividing, post-mitotic cells. • Specific serotypes (AAV2, AAV5, AAV9) heavily target neurons and retinal cells.
Episomal Expression	• Forms stable concatemers in the cell nucleus. • No chromosomal integration, leaving host DNA entirely untouched. • Near-zero risk of insertional oncogenesis.
Packaging Limit	• Strict 4.7 kb ceiling on genetic payload capacity. • Limits transgene size, preventing delivery of exceptionally large genes.

Clinical Hallmark: Delivers long-lasting gene expression in non-dividing neurons with remarkably low inflammatory immunogenicity.

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Lentiviral & Retroviral Vectors: Permanent Genomic Integration

[Vector: Viral] [Integration: Permanent] [Payload: 8–10 kb]

Stable Integration	• Permanently inserts the therapeutic transgene into host chromosomal DNA. • Ensures genetic transmission to all future daughter cells upon division.
Cargo Capacity	• Accommodates large payloads up to 8–10 kb. • Highly versatile for complex gene sequences and promoters.
Insertional Mutagenesis	• High risk of oncogenesis if integrated adjacent to proto-oncogenes. • Can disrupt normal cellular tumor-suppressor functions.



Clinical Niche: The primary tool for ex vivo hematopoietic stem cell therapy (ADA-SCID) and CAR-T cell oncology.

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Adenoviral Vectors & Alternative Viral Platforms

Adenovirus (AdV)

- **Massive capacity**: Holds up to 30 kb in helper-dependent gutless vectors.
- **Non-integrating**: Double-stranded DNA remains episomal.
- **High immunogenicity**: Capsid proteins trigger severe acute innate immune responses.
- **Best application:** Transient expression; ideal for vaccines rather than permanent cures.

Herpes Simplex Virus (HSV)

- **Enormous payload**: Exceeds 30 kb capacity for massive genetic constructs.
- **Naturally neurotropic**: Highly evolved to infect human nerve cells.
- **Persistent latency**: Can remain dormant in sensory ganglia for a lifetime.
- **Best application:** Complex neurological targeting where other viruses fail.

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Lipid Nanoparticles (LNPs): Non-Viral Nanotechnology

[Vector: Chemical] [Integration: None] [Payload: Highly Variable]

Molecular Architecture

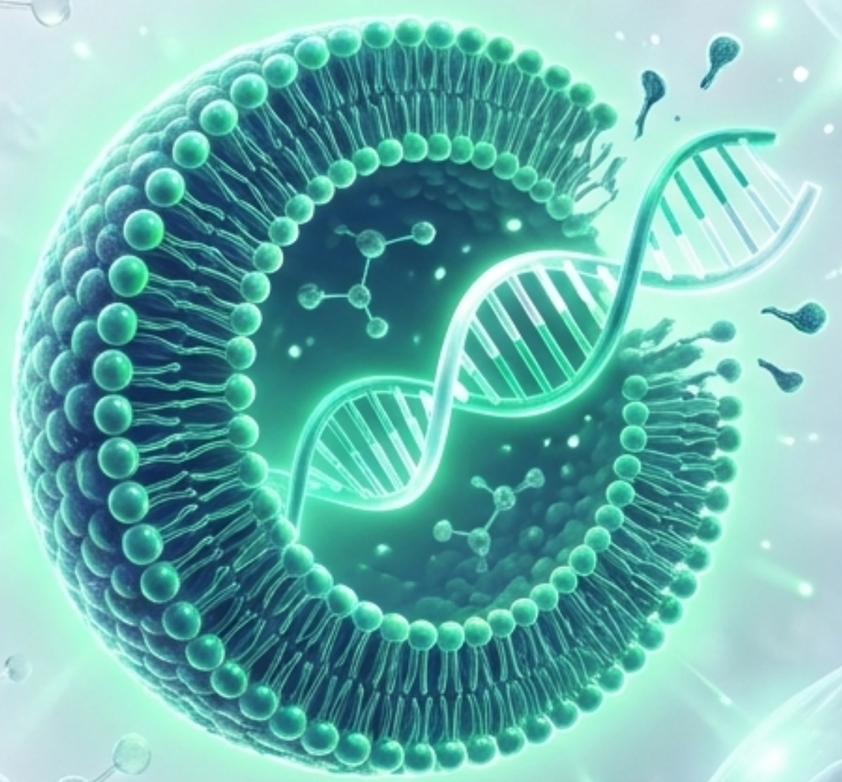
- **Ionizable cationic lipids**, **cholesterol**, and **PEG-lipids** form a protective sphere.
- Neutral at **physiological pH**, preventing toxic interactions in the blood.

Cellular Uptake & Escape

- Enters via endocytosis, becoming **positively charged** in the **acidic endosome**.
- **Ionizable lipids disrupt** the **endosomal membrane**.
- Facilitates **rapid release of therapeutic mRNA** into the cytoplasm.

Clinical Strengths

- No viral protein capsids, resulting in minimal adaptive immune neutralization.
- Allows repeated dosing without the body building viral immunity.
- Massive scalability for global manufacturing.



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Physical Delivery Modalities: Direct Membrane Entry

Electroporation

- **Short, high-voltage pulses** transiently destabilize the cell membrane.
- **Creates aqueous nanopores** allowing nucleic acids to rapidly slip inside.
- **Highly efficient** for ex vivo laboratory cell modifications.

Microinjection & Ballistic

- **Direct mechanical injection** into individual cells using micro-capillaries.
- **Ballistic bombardment** fires gold microparticles coated in DNA.
- **Labor-intensive** but **highly precise** for zygotes and early embryos.

Hydrodynamic Delivery

- **High-pressure fluid injection** into the bloodstream.
- **Transiently permeabilizes** capillary endothelia in targeted organs (like the liver).
- Primarily utilized in translational animal models.

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Neurodelivery: Penetrating the Blood-Brain Barrier (BBB)

The BBB Challenge

- Continuous **tight junctions** between brain capillary endothelial cells.
- Astrocytic **end-feet** and active **efflux transporters** provide secondary defense.
- Blocks **>98%** of systemic macromolecules from entering brain tissue.
- Renders standard **IV delivery** largely ineffective for CNS disorders.



Bypassing the Barrier

- **Intrathecal Lumbar Puncture**: Injects directly into **cerebrospinal fluid**, bathing the CNS.
- **Convection-Enhanced Delivery (CED)**: Uses a **pressure gradient** to push vectors through brain parenchyma.
- **Stereotactic Injection**: Surgical **intraparenchymal delivery** directly into specific deep brain structures.

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The Vector Synthesis Matrix

Vector System	Payload	Integration	Immunogenicity	Primary Use
AAV (Viral)	Very Low (4.7kb)	Episomal (Safe)	Low	In Vivo CNS/Retina
Lentivirus (Viral)	Medium (8-10kb)	Permanent	Low/Medium	Ex Vivo Stem Cells
Adenovirus (Viral)	High (up to 30kb)	Episomal	Very High (Acute)	Vaccines / Transient
LNPs (Non-Viral)	Variable/High	None	Very Low (Repeatable)	Systemic mRNA / Vaccines

Insight: Viral vectors trade payload capacity for evolutionary efficiency, while synthetic LNPs maximize safety and repeatability at the cost of targeting precision.

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Clinical Translation & Lesson Summary

Translating to F172 Target Diseases

Huntington's Disease (HD): Targeted via **AAV** or **Non-Viral Exosomes** directly to the striatum to silence mutant HTT.

Spinal Muscular Atrophy (SMA): Treated via **Antisense Oligonucleotides (Nusinersen)** delivered through **intrathecal lumbar puncture**.

ADA-SCID: Addressed via **ex vivo Lentiviral integration** into patient hematopoietic stem cells.

Core Takeaways

- **No 'Perfect" Vector:** Delivery choice is a strict compromise between **payload size**, **integration risk**, and **immune clearance**.
- **Anatomy Dictates Strategy:** The **Blood-Brain Barrier** forces CNS therapies toward localized in situ administration.
- **Future Trajectory:** The field is rapidly shifting from highly immunogenic **viral capsids** to infinitely scalable, **non-viral Lipid Nanoparticles**.

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